

Technology Stack

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	2 Marks

Technologies chosen for each layer of the Credit Card Fraud Detection solution.

S.No	Architecture Component	Technology Chosen	Justification
1	Frontend / Client-Side	HTML, CSS, Bootstrap (Flask Jinja templates)	Lightweight, quick to build a simple analyst-facing input form
2	Backend / Server-Side	Python, Flask	Fast to prototype, integrates directly with scikit-learn/XGBoost models
3	Modeling / ML Libraries	scikit-learn, XGBoost, pandas, numpy, imbalanced-learn (SMOTE)	Industry-standard libraries for classical ML + gradient boosting + imbalance handling
4	Database / Data Storage	CSV dataset (creditcard.csv); optionally SQLite for logging predictions	Dataset is provided as CSV; SQLite is simple for a small-scale demo
5	Cloud / Hosting / Deployment	IBM Watson Machine Learning, Ngrok (for local demo tunneling)	Enables scalable, real-time hosted scoring and easy public demo access
6	Version Control	GitHub	Team collaboration, versioning, and submission requirement
7	Third-Party APIs / Other Tools	IBM Watson ML REST API	Used to deploy and serve the trained model for real-time predictions